

Superscalar Processors

- Increasing pipeline length eventually leads to diminishing returns
 - longer pipelines take longer to re-fill
 - data and control hazards lead to increased overheads, removing any performance advantage
 - yet there is still space for more circuitry on-chip
 - Feature size still falling
 - Improved purity in crystals means larger die size possible
- An alternative is the *multiple-issue processor*
 - i.e. issue more than one instruction each clock cycle
 - that is, start more than one instruction
- Such processors are called **Superscalar** processors

Copyright 1998, 2002 © Leslie S. Smith

31R6 - Computer Design

Slide 82

Characteristics of superscalar processors

- *Superscalar* processors
 - issue more than one instruction each cycle
 - the number of instructions issued will depend on the instructions in the instruction stream
 - instructions are often re-ordered to fit the processor architecture better
 - typically, between 1 and 8 instructions can be issued each cycle
- The simplest possibility is a machine which issues
 - one integer instruction
 - one floating-point instruction
 - per clock cycle
- ...but this only gives appreciable improvements if there are many FP instructions to be executed.

Copyright 1998, 2002 © Leslie S. Smith

31R6 - Computer Design

Slide 83

Ordering instructions 1

- Instructions are produced in a particular order by the compiler, but need not (necessarily) be executed in that order.
- They may be
 - issued in order and completed in order
 - issued in order, but completed out of order
 - issued out-of-order, but completed in order
 - issued out of order and completed out of order.

Copyright 1998, 2002 © Leslie S. Smith

31R6 - Computer Design

Slide 84

Ordering instructions 2

- The best order for instructions in a particular superscalar architecture depends on
 - the architecture itself
 - the precise dependencies between instructions
- the actual order they are executed in may be
 - set up by the compiler
 - in which case it must “know” the architecture
 - complex code-generation phase in compiler
 - Can be very effective
 - or the instructions may be reordered by the CPU itself
 - many instructions are fetched, but the precise order of execution is decided at execution time
 - this is more common (currently)

Copyright 1998, 2002 © Leslie S. Smith

31R6 - Computer Design

Slide 85

Register renaming

- One of the main techniques for dealing with data hazards is register renaming
 - registers are dynamically allocated by the hardware
 - out of a register set which is larger than the programmer's model of the architecture
 - subsequently executing instructions use whichever register is correct for them.
 - Usually the actual register which a programmer's register refers to is re-allocated from time to time
- Limits to multiple issue machines
 - there are inherent limits to the extent to which sequential instructions can be executed concurrently
 - these show up as data and control hazards
 - there are limits to the amount of underlying hardware that can be produced
 - these show up as structural hazards
 - can better compilers help?

Copyright 1998, 2002 © Leslie S. Smith

31R6 - Computer Design

Slide 86

Branch Prediction

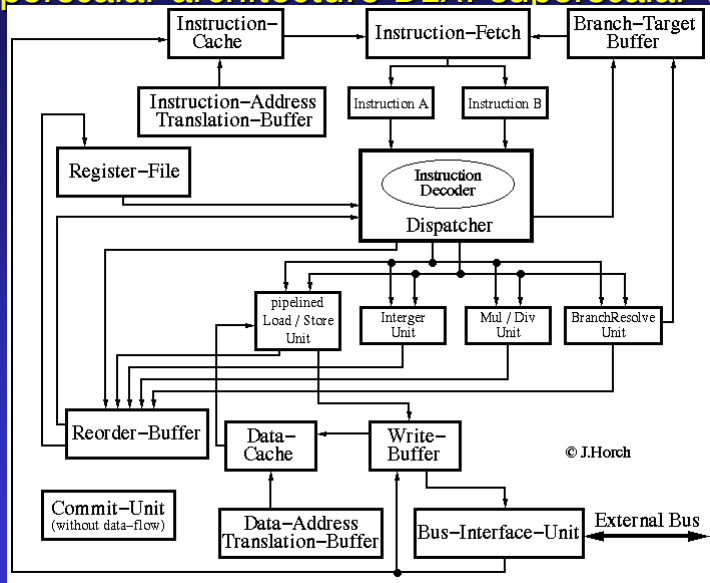
- Since branches can cause major delays in pipelines, some machines attempt to predict what the outcome of a branch will be
- Like the 68040, the Pentium attempts to limit the problems which branches cause by fetching instructions from both the target address and the branch address
- However, it predicts which will occur
 - from what has happened previously
- and (speculatively) executes from that address.
- Other techniques can use the compiler to predict whether the branch is more or less likely to be taken

Copyright 1998, 2002 © Leslie S. Smith

31R6 - Computer Design

Slide 87

Superscalar architecture DLX: superscalar version



Copyright 1998, 2002 © Leslie S. Smith

31R6 - Computer Design

Slide 88

Superscalar DLX Features

- - Pipelined, superscalar
 - Two instructions per clock
 - Branch-Target buffer
 - Reorder-Buffer to commit instructions in program order
 - Precise exception processing
- Four execution units with Reservation-Station
 - Branch-Resolve unit
 - Arithmetic-Logic unit
 - Multiply-Divide unit
 - Load-Store unit
- Write-Buffer

Copyright 1998, 2002 © Leslie S. Smith

31R6 - Computer Design

Slide 89

Example 2: DEC ALPHA AXP 21064

- DEC ALPHA AXP 21064
 - superscalar architecture with
 - multiple functional units
 - 7 stage 5ns/stage integer pipeline
 - the CPU can issue simultaneously
 - any load/store in parallel with any operate
 - an integer operate in parallel with a FP operate
 - a floating operate and a floating branch
 - an integer operate and an integer branch
 - the most recent version of the ALPHA can issue 4 instructions simultaneously.
 - 4 integer execution units
 - 2 FP execution units

Copyright 1998, 2002 © Leslie S. Smith

31R6 - Computer Design

Slide 90

Pentium architecture

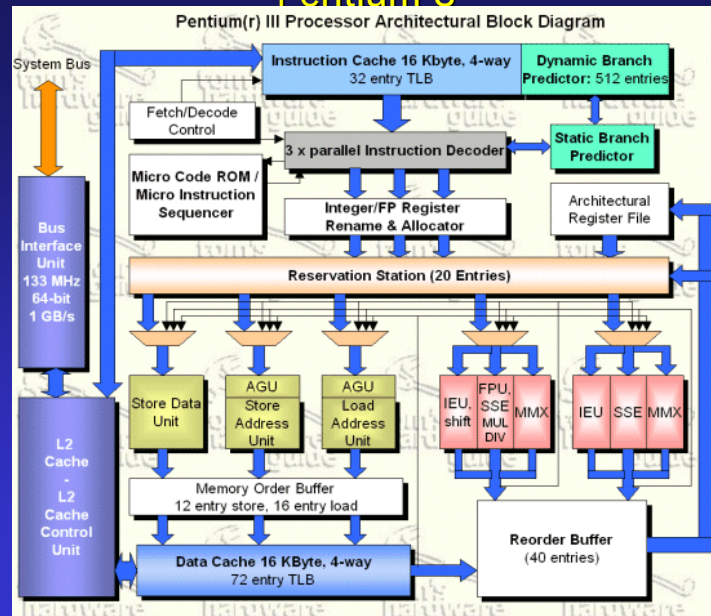
- Superscalar architecture
 - 2 independent integer pipelines
 - one Floating point pipeline
 - but control unit can issue either 2 integer instructions or 1 (occasionally 2) floating point instructions
- Branch prediction by loading code to cache from target address
- separate on-chip instruction and data caches
 - both 8Kbytes long
- Integer pipelines are not identical
 - u pipeline includes barrel shifter, but v is simpler (and can't execute all integer instructions)
- Some elements of microcoded control using control ROM.

Copyright 1998, 2002 © Leslie S. Smith

31R6 - Computer Design

Slide 91

Pentium 3



Copyright 1998, 2002 © Leslie S. Smith

31R6 - Computer Design

Slide 92

Pentium 3

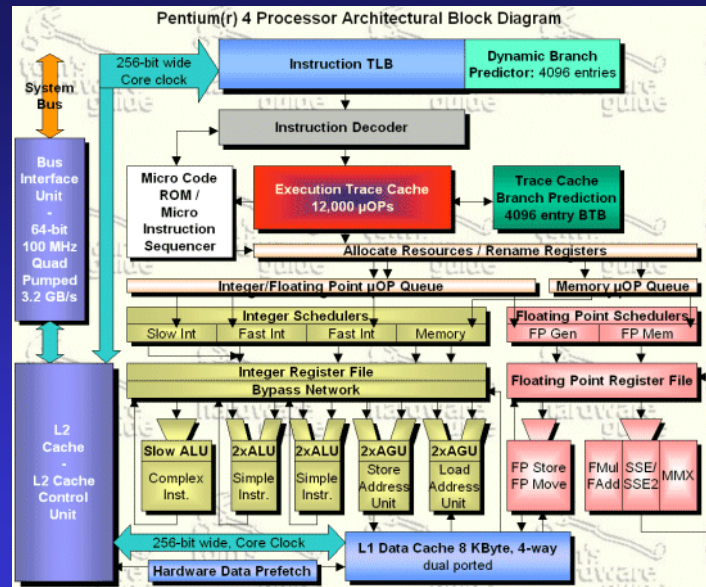
- Superscalar processor
 - 3 integer pipelines
 - 2 FP pipelines
- 16KB instruction cache
- 16KB data cache
- L2 cache on chip
- Microcode system to deal with CISC instructions
 - some x86 instructions -> 1 microOp
 - others into 2 or 3 microOps
 - others into many more

Copyright 1998, 2002 © Leslie S. Smith

31R6 - Computer Design

Slide 93

Pentium 4



Copyright 1998, 2002 © Leslie S. Smith

31R6 - Computer Design

Slide 94

Pentium P4

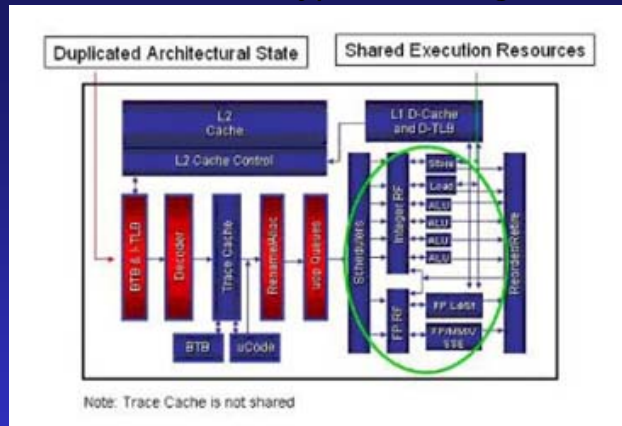
- Similar to P3, but designed for ultra-high clock rates
- Quad-pumped (“pipelined”) system bus
- 20 stage pipeline (!)
 - P3 has 10
- Sophisticated branch prediction using BTB
 - crucial for performance with long pipelines.
- Ultrafast integer ALU (2xclock speed) for simple integer operations
- Execution trace Cache of uOPs
 - instead of Instruction cache
- Smaller Data cache
 - to allow 1-cycle access
- New FP instructions
 - but is anyone using them?

Copyright 1998, 2002 © Leslie S. Smith

31R6 - Computer Design

Slide 95

Hyperthreading



Hyperthreading means sharing the execution resources between a number (normally 2) of sets of execution resources.

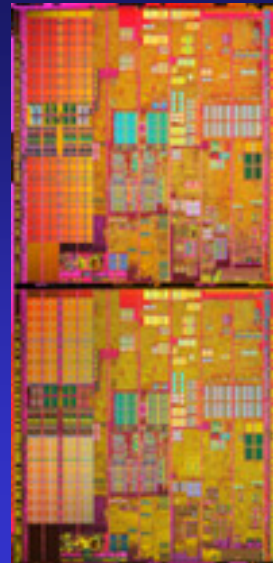
Copyright 1998, 2002 © Leslie S. Smith

31R6 - Computer Design

Slide 96

Dual Core

- Dual core processors essentially have two complete processors on one silicon chip
- This allows an essentially dual processor machine to be built cheaply
- Sometimes (as in the Intel case) some caching (specifically L2 cache) is shared between the processors
- Note that this can be mixed with hyper threading



Copyright 1998, 2002 © Leslie S. Smith

31R6 - Computer Design

Slide 97